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(a)(i)	The graph is a straight line, which indicates that force (and hence acceleration, since mass is constant) is proportionate to displacement
	from the equilibrium position of 0.85 m.
	The graph has negative gradient, which indicates that force (and hence acceleration) points towards an equilibrium position.
(a)(ii)1.	Amplitude = $1.20 - 0.85$ or $0.85 - 0.50 = 0.35$ m
(a)(ii)2.	$a_0 = \omega^2 x_0$ $\omega = \sqrt{\frac{a_0}{x_0}} = \sqrt{\frac{3.42}{0.35}}$ $= 3.1259 \text{ rad s}^{-1}$
	$f = \frac{\omega}{2\pi} = 0.4975 \text{ Hz}$
	The building's oscillation is driven by the wind, so its oscillation follows the frequency of the wind
(a)(iii)	
	Correct shape of either graph
	Correct shape of other graph, and correct x-values

(a)(iv)	$\frac{KE}{PE} = \frac{1}{2} \Rightarrow \frac{KE}{KE_{\max}} = \frac{KE}{KE + PE} = \frac{1}{3}$ $\frac{\frac{1}{2}m(\omega\sqrt{c_0^2 - c^2})^2}{\frac{1}{2}m(\omega c_0)^2} = \frac{1}{3}$
	$\frac{c_0^2 - c^2}{c_0^2} = \frac{1}{3}$ $\frac{c^2}{c_0^2} = \frac{2}{3}$ $c^2 = \frac{2}{3}(0.35)^2$ $c = 0.29m$ $x = 0.29 + 0.85 = 1.14m$
(b)	The natural frequency of those buildings matched the frequency of the earthquake, so resonance occurred.
	There is maximum transfer of energy to the buildings / the buildings oscillate with maximum amplitude

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5(a)

The principle of superposition states that the net displacement at a given place and time